



Hemolytic anemias

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Hemolytic anemias

- Are defined as anemias that result from an increase in the rate of red cell destruction.
- Because of erythropoietic hyperplasia and anatomical extension of bone marrow, red cell destruction may be increased several-fold before the patient becomes anemic – compensated hemolytic disease.
- The normal adult marrow, after full expansion, can produce red cells at 6–8 times the normal rate.
- **It leads to a marked reticulocytosis.**
- Therefore, anemia due to hemolysis may not be seen until the red cell lifespan is less than 30 days.

Table 6.1 Classification of haemolytic anaemias.

Hereditary

Membrane

Hereditary spherocytosis, hereditary elliptocytosis

Metabolism

G6PD deficiency, pyruvate kinase deficiency

Haemoglobin

Genetic abnormalities (Hb S, Hb C, unstable); see Chapter 7

Acquired

Immune

Autoimmune

Warm antibody type (see Table 6.5)

Cold antibody type

Alloimmune

Haemolytic transfusion reactions

Haemolytic disease of the newborn

Allografts, especially marrow transplantation

Drug associated

Red cell fragmentation syndromes

See Table 6.6

March haemoglobinuria

Infections

Malaria, clostridia

Chemical and physical agents

Especially drugs, industrial/domestic substances, burns

Secondary

Liver and renal disease

Paroxysmal nocturnal haemoglobinuria (see Chapter 22)

G6PD, glucose-6-phosphate dehydrogenase; Hb, haemoglobin.

- As the RBC cells have no nucleus, and their metabolism gradually deteriorates as enzymes are degraded, the cells become non-viable.
- The breakdown of hemoglobin from hemoglobin liberates iron from recirculation via plasma transferrin mainly to marrow erythroblasts, and protoporphyrin, broken down to **bilirubin**.
- Bilirubin circulates to the liver where it is conjugated to glucuronides, which are excreted into the gut via bile and converted to **stercobilinogen and stercobilin**.
- Stercobilinogen and stercobilin are partly reabsorbed and excreted in urine as **urobilinogen and urobilin**. Globin chains are broken down into amino acids and reutilized for general protein synthesis in the body.
- **Haptoglobins** are proteins in normal plasma that bind hemoglobin. The reticuloendothelial system (RE) system removes the hemoglobin-haptoglobin complex. Intravascular hemolysis (breakdown of red cells within blood vessels) plays little or no part in normal red cell destruction.

Intravascular and extravascular hemolysis

There are two mechanisms whereby red cells are destroyed in hemolytic anemia.

1. There may be excessive removal of red cells by cells of the RE system (**extravascular hemolysis**).
2. **or they** may be broken down directly in the circulation (**intravascular hemolysis**) **Which every mechanism dominates** will depend on the pathology involved.

- In intravascular hemolysis, free hemoglobin is released which rapidly saturates plasma haptoglobins and the excess free hemoglobin is filtered by the **glomerulus**.
- If the hemolysis rate saturates the renal tubular re-absorptive capacity, free hemoglobin enters the urine.
- Iron released from hemoglobin in the renal tubules is seen as haemosiderin in a urinary deposit.
- Methaemalbumin is also formed from the process of intravascular hemolysis.

The main laboratory features of intravascular hemolysis therefore are:

- 1. Haemoglobinaemia and hemoglobinuria**
- 2. Haemosiderinuria**
- 3. Methaemalbuminaemia**

Extravascular Hemolytic Anemia

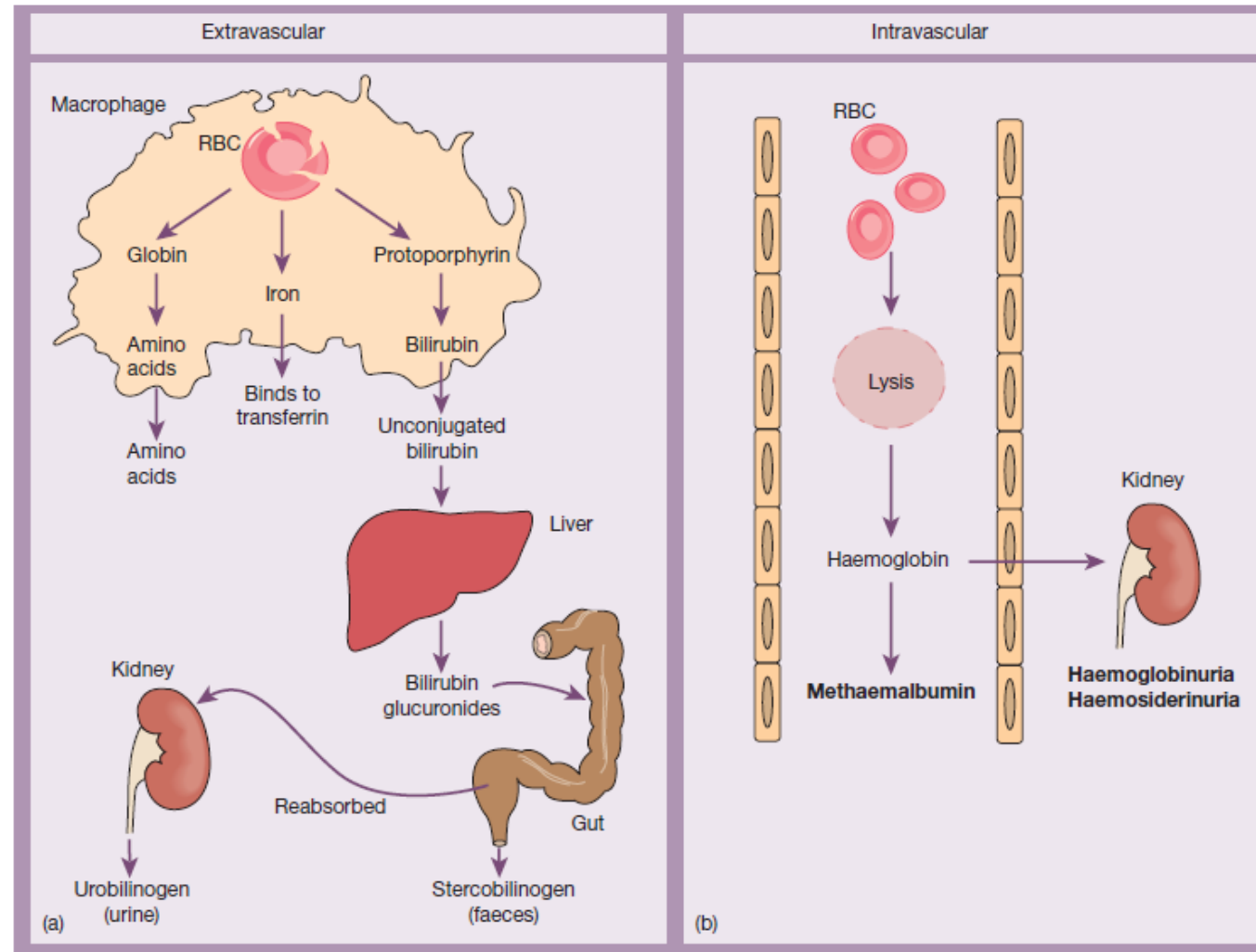
- Many hemolytic anemias result from increased extravascular hemolysis, during which more than the usual number of RBCs are removed from the circulation daily.
- If the number of destroyed RBCs is more than that produced from the BM, the anemia develops.
- The destruction of RBCs occurs in the macrophages of the spleen and liver (extravascular hemolysis).
- Extravascular hemolysis is associated with the production of spherocytes.
- Spherocytes result from the ingestion of a portion of the RBC membrane.
- The plasma bilirubin level is increased, with largely unconjugated.

Intravascular hemolytic anemia

- In intravascular hemolytic anemia, RBCs are destructed much more than the daily amount.
- It causes a profound anemia.
- It results in fragmentation of RBCs that release detectable Hb into the plasma.
- It is characterized by: hemoglobinemia, hemoglobinuria, and hemosiderinuria.
- Haptoglobin level is decreased.
- Fragmented RBCs (schistocytes) are noted.
- The plasma bilirubin level is increased, with largely unconjugated.

Normal red cell destruction

- With no nucleus RBCs gradually deteriorate (120 days lifespan), and enzymes are degraded and not replaced.



Clinical features

- Fatigue
- Dyspnea
- Dizziness
- Pallor of the mucous membrane
- Tachycardia
- Jaundice (color of the skin and sclera)
- Splenomegaly can develop
- Gallstones can occur whenever hemolysis is chronic
- Urine turns dark (excess urobilinogen).

General lab findings of hemolytic anemias

1. Features of increased red cell breakdown:
 - Increase serum bilirubin (both unconjugated and bound to albumin)
 - Increase urine urobilinogen
2. Features of increased red cell production:
 - Reticulocytosis.
 - Erythroid hyperplasia (myeloid erythroid ratio reduced)
3. Damaged red cell features:
 - Morphology (micro-spherocytes, elliptocytes, fragments)
 - Osmotic fragility, autohaemolysis, etc.

Laboratory findings

1. Tests of Accelerated RBC Destruction:

- Bilirubin:↑ (largely unconjugated)
- LDH:↑(more in intravascular hemolysis)
- Haptoglobin:↓ (intravascular hemolysis)

2. Tests of increased RBC Production (erythropoiesis):

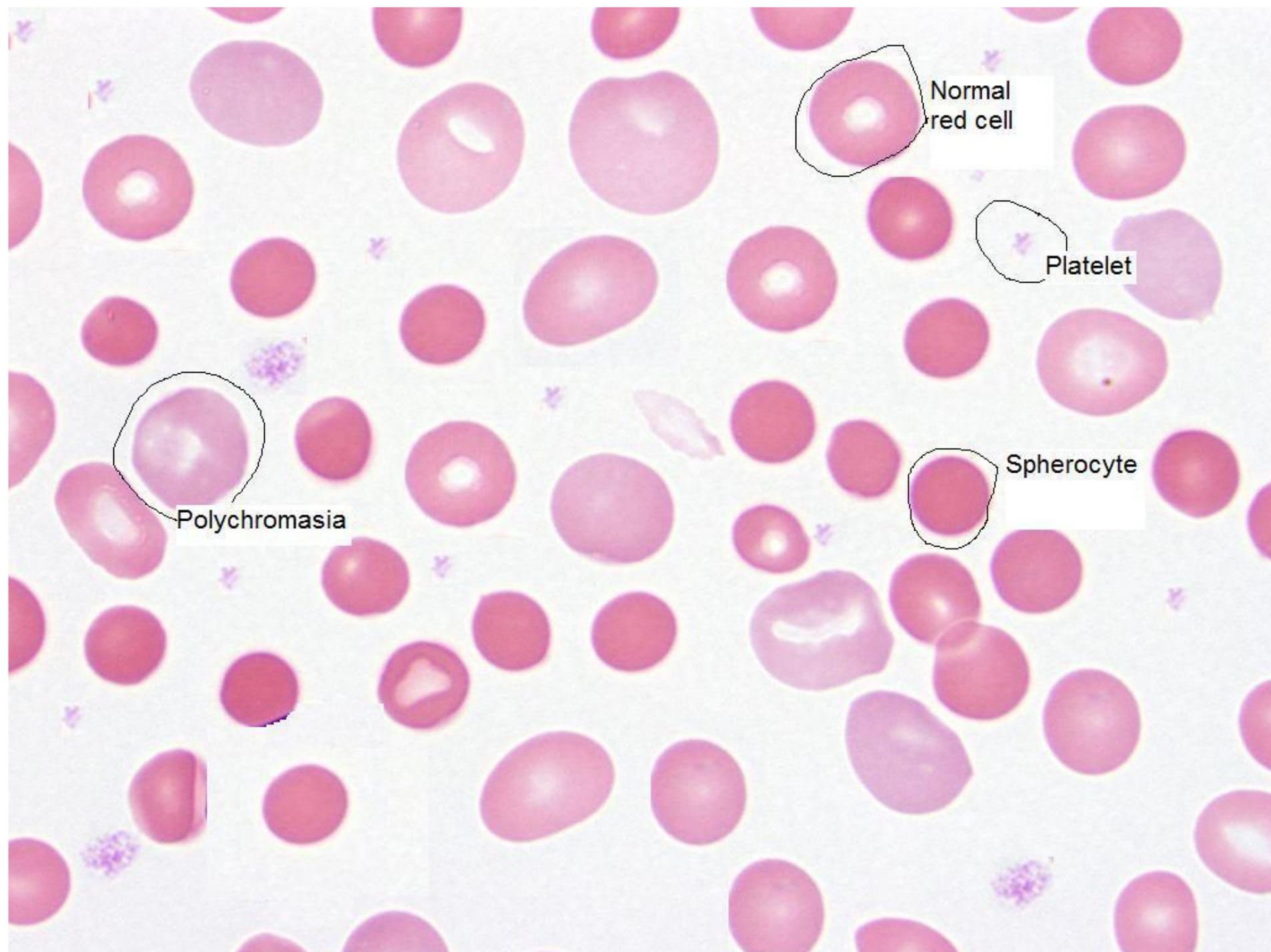
- Retics:↑ (Reticulocytosis)
- Rising MCV due to reticulocytosis, compared with value seen early in hemolysis.
- BM biopsy: Erythroid hyperplasia (not necessary)

3. Blood film:

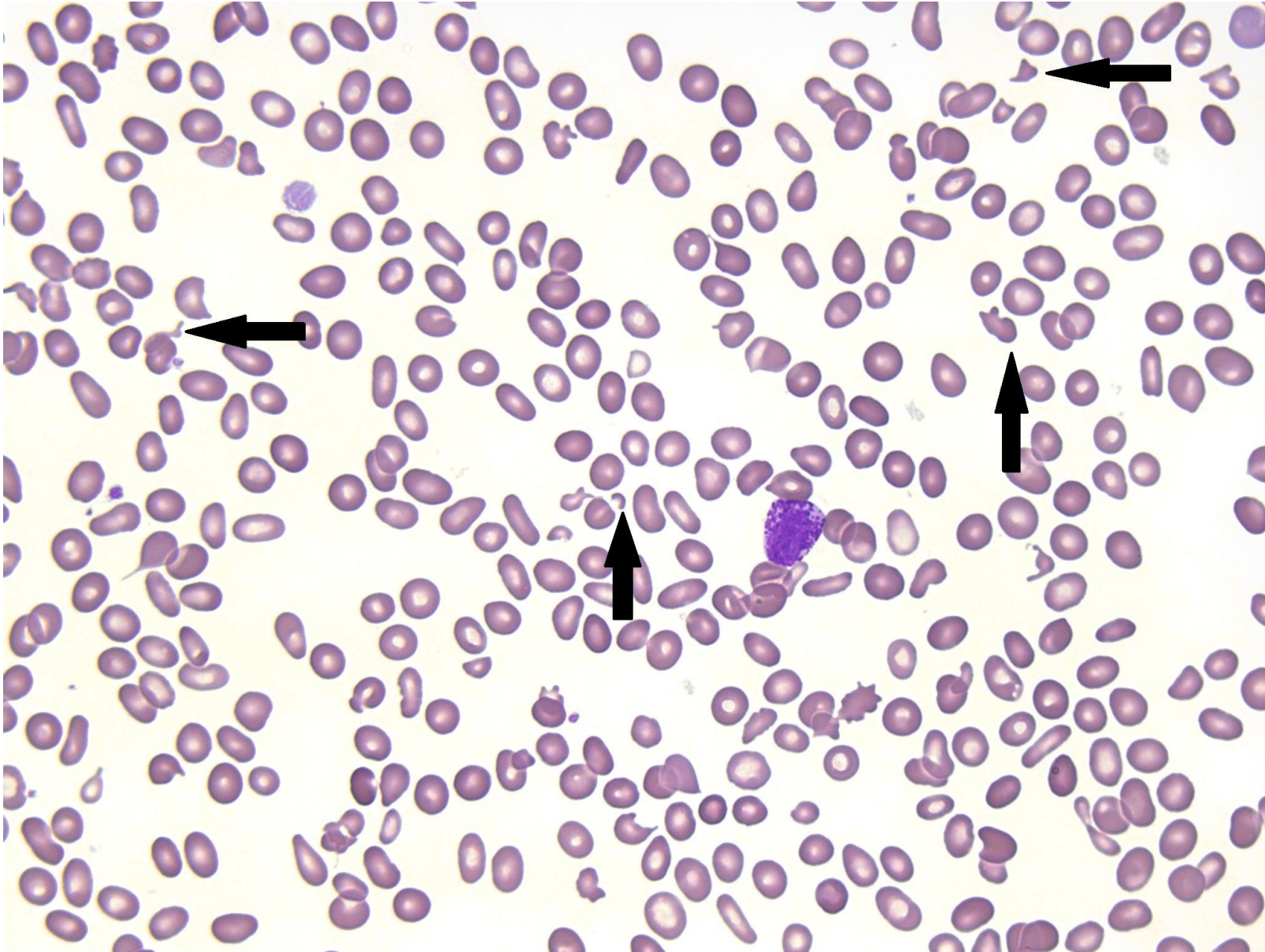
- Polychromasia
- Nucleated RBCs
- Spherocytes (extravascular hemolysis)
- Schistocytes (intravascular hemolysis)

4. In all types of hemolytic anemias:

- Hb, RBCs count: ↓
- Bilirubin, LDH, Retics: ↑
- MCV: raised compared with the value seen early in hemolysis.
- Polychromasia
- Nucleated RBCs



Schistocytes



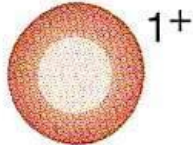
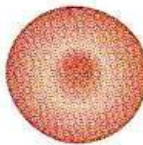
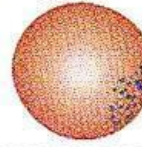
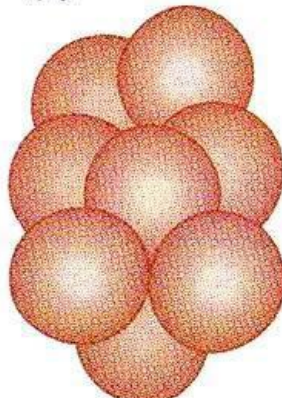
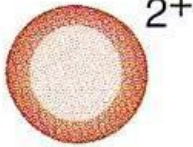
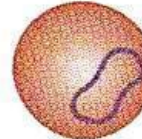
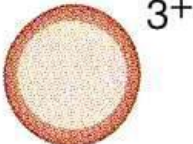
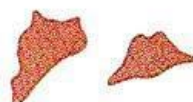
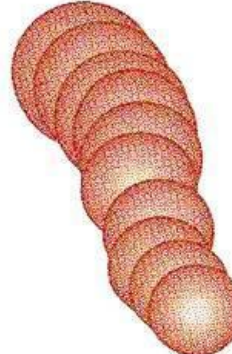
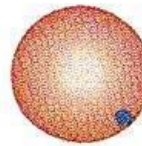
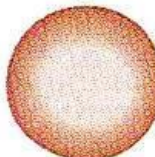
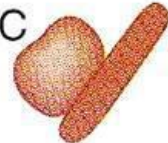
Laboratory tests to determine specific hemolytic processes:

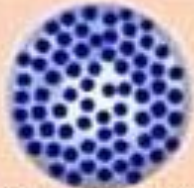
- Well-stained blood film (sickle cell, elliptocytes (ovalocytes), spherocytes, schistocytes, acanthocytes)
- Osmotic fragility test (hereditary spherocytosis)
- Direct Antiglobulin Test (DAT): to differentiate between immune and non-immune hemolytic anemias.
- Hb electrophoresis (hemoglobinopathies)
- Enzyme assays (G6PD)

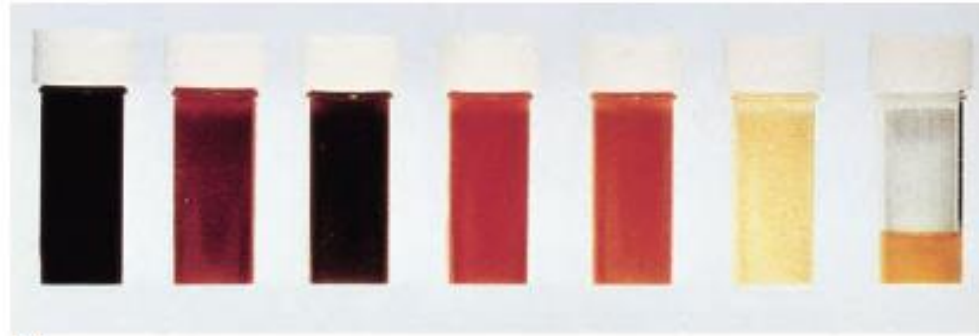
Differential Diagnosis:

- Intravascular vs. extravascular.
- Hemolytic anemia should be differentiated from other anemias with bilirubinemia, reticulocytosis, or both.
- Some anemias have reticulocytosis (recovery from hemorrhage, after treatment with iron, B12, and folic acid).
- Megaloblastic anemia is a hemolytic due to ineffective erythropoiesis (LDH and bilirubin ↑), but the Retics are **low**.
- Megaloblastic anemia is anemia of decreased production of RBCs as most of the RBCs never reach the peripheral blood.

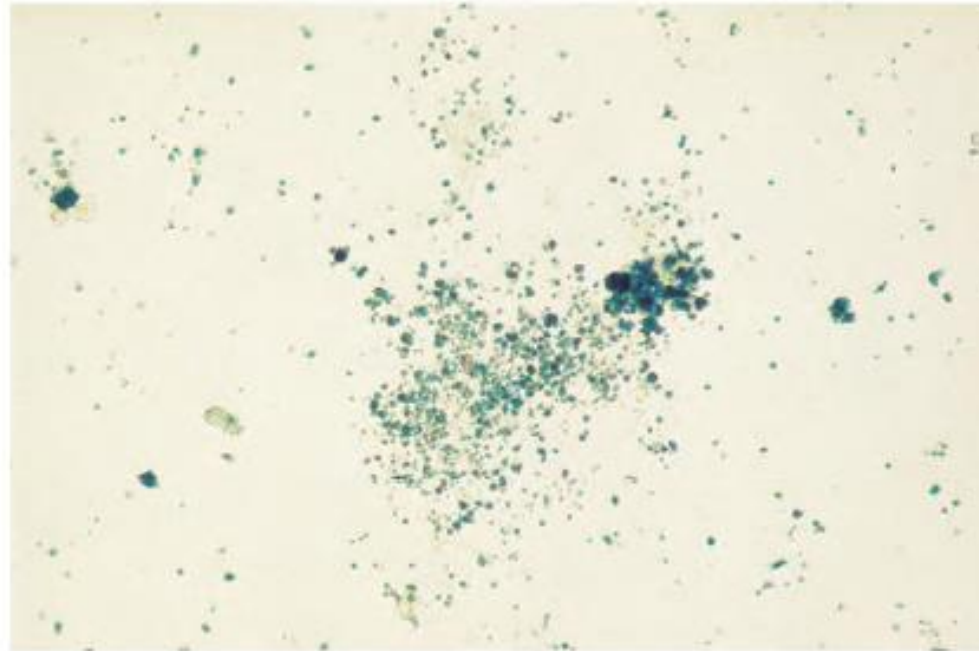
RED BLOOD CELL MORPHOLOGY

Size variation	Hemoglobin distribution	Shape variation		Inclusions	Red cell distribution
Normal 	Hypochromia 1+ 	Target cell 	Acanthocyte 	Pappenheimer bodies (siderotic granules) 	Agglutination 
Microcyte 	 	Spherocyte 	Helmet cell (fragmented cell) 	Cabot's ring 	
Macrocyte 	 	Ovalocyte 	Schistocyte (fragmented cell) 	Basophilic stippling (coarse) 	Rouleaux 
Oval macrocyte 	 	Stomatocyte 	Tear drop 	Howell-Jolly 	
Hypochromic macrocyte 	Polychromasia  (Reticulocyte)	Sickle cell 	Burr cell 	Crystal formation	
				HbSC 	HbC 

Common RBC Inclusions	Cartoon Image	Inclusion	May be associated with
Howell Jolly Bodies		DNA	Hyposplenism Asplenism Severe hemolytic anemia
Heinz Bodies	 <i>Supravital stain</i>	Hemoglobin	G6PD deficiency Oxidant drugs Unstable hemoglobin
Pappenheimer Bodies		Iron deposits	Thalassemia Sideroblastic anemia Hemolytic anemia Post-splenectomy
Hemoglobin H Inclusion	 <i>Supravital stain</i>	Hemoglobin	Hemoglobin H disease
Basophilic Stippling		Ribosomes	Lead poisoning Thalassemia Sickle cell anemia MDS



(a)



(b)

Figure 6.3 (a) Progressive urine samples in an acute episode of intravascular haemolysis showing haemoglobinuria of decreasing severity. (b) Prussian blue-positive deposits of haemosiderin in a urine spun deposit (Perls' stain).

Hereditary hemolytic anemias

I. Membrane defects

A. Hereditary spherocytosis (HS):

Hereditary spherocytosis (HS) is the most common hereditary hemolytic anemia in Northern Europeans.

Pathogenesis:

- HS is usually caused by defects in the proteins involved in the vertical interactions between the membrane skeleton and the lipid bilayer of the red cell.
- The marrow produces red cells of normal biconcave shape but these lose membrane and become increasingly spherical (loss of surface area relative to volume) as they circulate through the spleen and the rest of the RE system.
- The loss of membrane may be caused by the release of parts of the lipid bilayer that are not supported by the skeleton.
- Ultimately, the spherocytes are unable to pass through the microcirculation, especially in the spleen, and die prematurely.

Table 6.3 Molecular basis of hereditary spherocytosis and elliptocytosis.

Hereditary spherocytosis

Ankyrin deficiency or abnormalities

α - or β -spectrin deficiency or abnormalities

Band 3 abnormalities

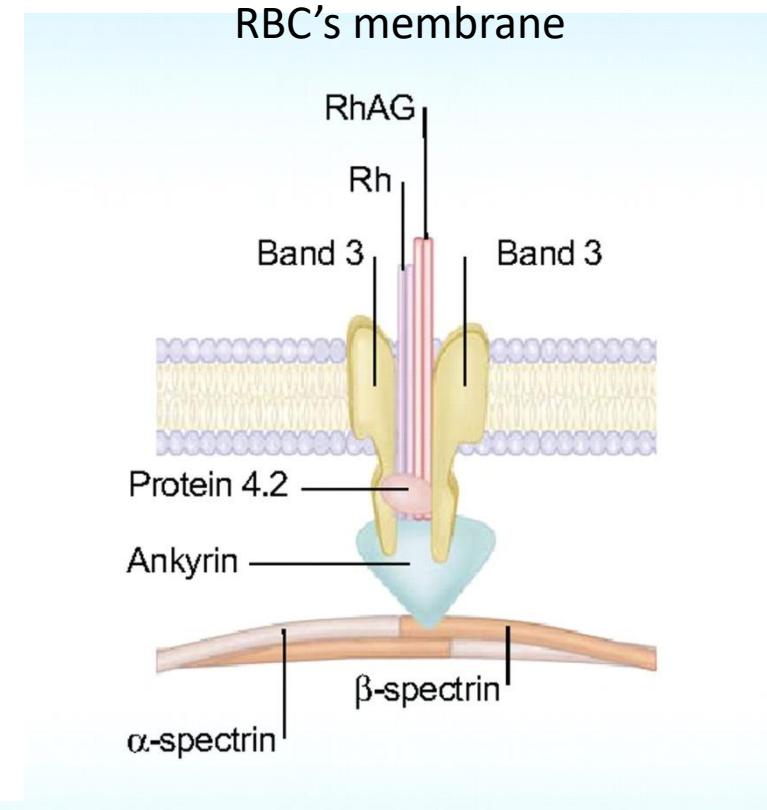
Pallidin (protein 4.2) abnormalities

Hereditary elliptocytosis

α - or β -spectrin mutants leading to defective spectrin dimer formation

α - or β -spectrin mutants leading to defective spectrin–ankyrin associations

Protein 4.1 deficiency or abnormality



Clinical features

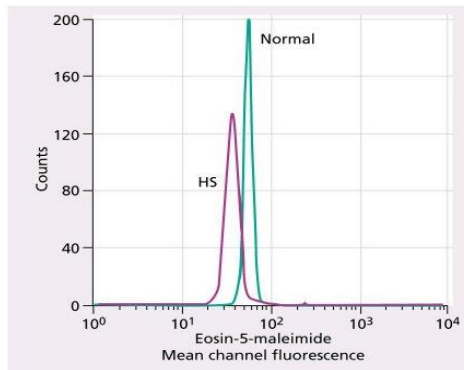
- Jaundice is typically fluctuating and is particularly marked if the hemolytic anemia is associated with Gilbert's disease (a defect of hepatic conjugation of bilirubin).

Hereditary spherocytosis lab findings:

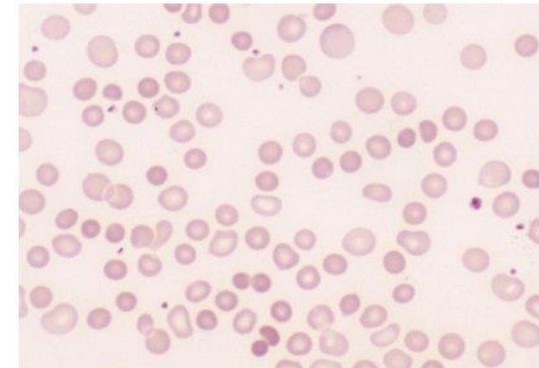
- Reticulocytes increase 5-20% (normal 0.5-2.5%) due to increased erythropoiesis (erythroid hyperplasia).
- Blood film (spherocyte mixed with normal RBCs, abundant reticulocytes (large polychromatic cells)).

Special tests include:

1. Rapid fluorescent flow analysis of eosin-maleimide (EMA) bound to the red cell. (decrease). The EMA test has replaced the osmotic fragility test.
2. Direct antiglobulin test (coombs test) normal (to exclude autoimmune cause).

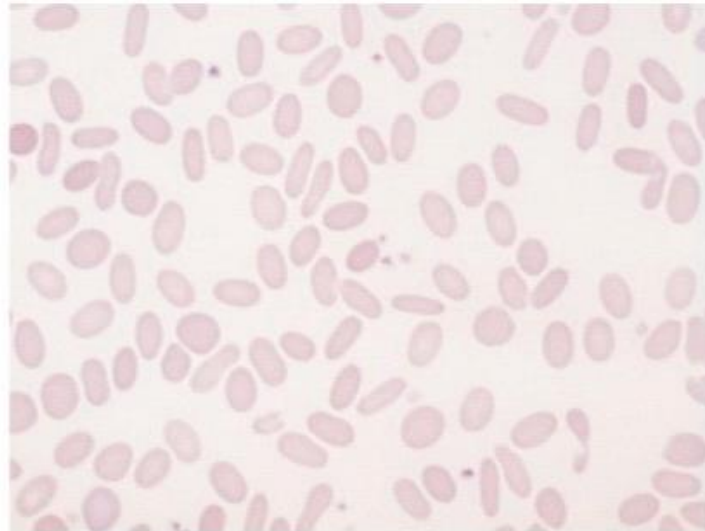


Splenectomy is the treatment



B. Hereditary elliptocytosis (HE):

- This has similar clinical and laboratory features to HS except for the appearance of the blood film
- Usually a clinically milder disorder.
- It is predominantly discovered by chance on a blood film and there may be no evidence of hemolysis.
- Occasional patients require splenectomy.
- **The basic defect is a failure of spectrin heterodimers to self-associate into heterotetramers.**



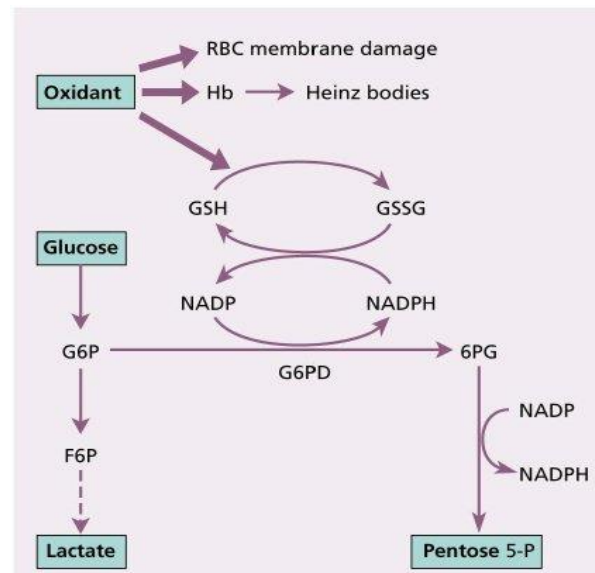
Elliptocyte RBCs

II. defective red cell metabolism:

1. G6PD deficiency:

- Function to reduce NADPH.
- NADPH is needed for the production of glutathione.
- Glutathione in the cells is used to mop up free radicals that cause oxidative damage.
- People with G6PD deficiency are therefore at risk of hemolytic anemia in states of oxidative stress.

Oxidative stress can result from infection and chemical exposure to medication and certain foods. Broad beans, e.g., fava beans.



- 400 variants caused by point mutation or deletions.
- The inheritance is sex-linked affecting males and carried by females.
- G6PD deficiency is usually asymptomatic.
- G6PD is present in all cells but why affecting RBC?
- The G6PD / NADPH pathway is the *only* source of reduced glutathione in red blood cells (erythrocytes). The role of red cells as oxygen carriers puts them at substantial risk of damage from oxidizing free radicals except for the protective effect of G6PD/NADPH/glutathione.
- Symptoms (occurrence of the syndrome) appears as follows:
 1. Acute hemolytic anemia in response to oxidant stress
 2. Neonatal Jaundice

Table 6.4 Agents that may cause haemolytic anaemia in glucose-6-phosphate dehydrogenase (G6PD) deficiency.

Infections and other acute illnesses (e.g. diabetic ketoacidosis)

Drugs

Antimalarials (e.g. primaquine, pamaquine, chloroquine, Fansidar[®], Maloprim[®])

Sulphonamides and sulphones (e.g. co-trimoxazole, sulfanilamide, dapsone, Salazopyrin[®])

Other antibacterial agents (e.g. nitrofurans, chloramphenicol)

Analgesics (e.g. aspirin), moderate doses are safe

Antihelminths (e.g. β -naphthol, stibophen)

Miscellaneous (e.g. vitamin K analogues, naphthalene (mothballs), probenecid)

Fava beans (possibly other vegetables)

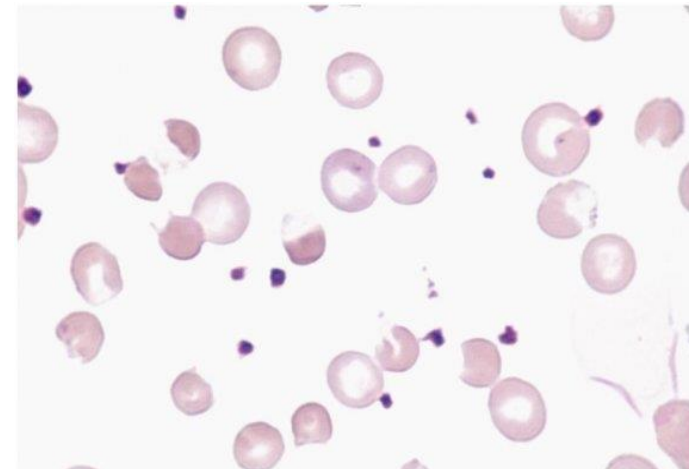
NB. Many common drugs have been reported to precipitate haemolysis in G6PD deficiency in some patients (e.g. aspirin, quinine and penicillin) but not at conventional dosage.

Diagnosis of G6PD:

- Between or without crisis blood cell count is normal (everything normal).
- During crisis blood film shows:
 - Loss of cytoplasm with separation of remaining hemoglobin from the cell membrane (blister cells).
 - Contracted and deeply stained cells (bite cells).
 - Oxidized denatured hemoglobin (Heinz bodies) may be seen in the reticulocytes.

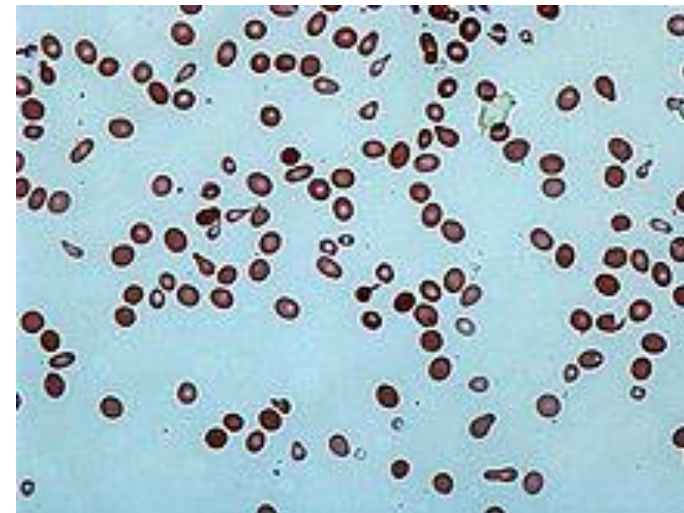
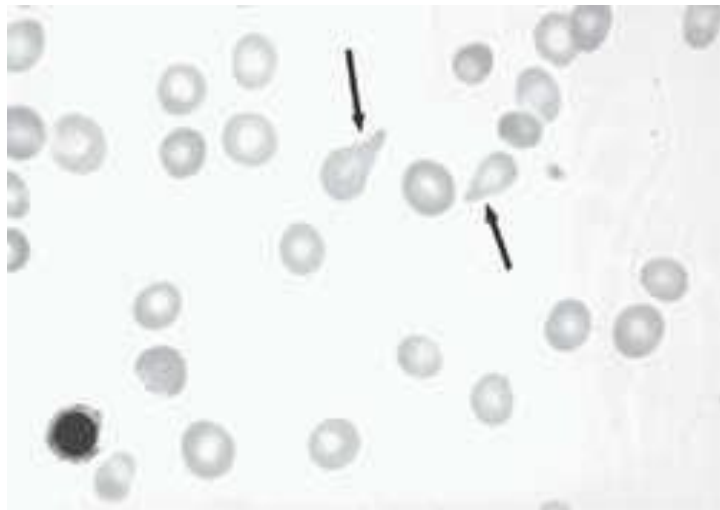
Treatments:

- Cause is stopped (drug, food, etc.)
- Blood transfusion may be considered.
- Baby jaundice treated by phototherapy or transfusion in severe cases.



2. Pyruvate kinase deficiency:

- This is inherited as an autosomal recessive, the affected patients being homozygous or doubly heterozygous.
- Over 100 different mutations have been described.
- The red cells become rigid as a result of reduced adenosine triphosphate (ATP) formation.
- The severity of the anemia varies widely (hemoglobin 40–100 g/L) and causes relatively mild symptoms.
- A rise in intracellular 2,3-diphosphoglycerate (2,3-DPG).
- Clinically, jaundice is usual and gallstones are frequent.
- The blood film shows poikilocytosis.
- Direct enzyme assay is needed to make the diagnosis.



Acquired hemolytic anemia

Immune hemolytic anemia:

1. Autoimmune hemolytic anemia (AIHA): antibody production against RBCs. **+ve Coomb's test (+ve DAT direct antiglobin test)** (agglutination of washed RBCs after adding anti-human globulin reagent).
 - Two types of autoimmune Hemolytic anemia
 - Warm autoimmune AIHA: usually IgG (react at 37C)
 - Cold AIHA: usually IgM type (react at 4C)
2. Alloimmune hemolytic anemia (caused by ABO-incompatible transfusion or Rh disease of the newborn).
3. Drug-induced immune hemolytic anemia. (Ab produced against a drug that binds to RBCs. Overdose of penicillin or deposition of complement via a drug-protein ex. Quinidine).